

# WEEK 8: Interfaces in Java

**Concepts Covered**

Defining Interfaces • Implementing Interfaces • Interface References • Extending Interfaces

## 1. PROBLEM STATEMENT

**Sample Program: Interfaces (CO2)**

Different categories of employees are working in a software company like **Regular Employees**, **Contract Employees**, and **Vendors**. Their payroll calculations differ:

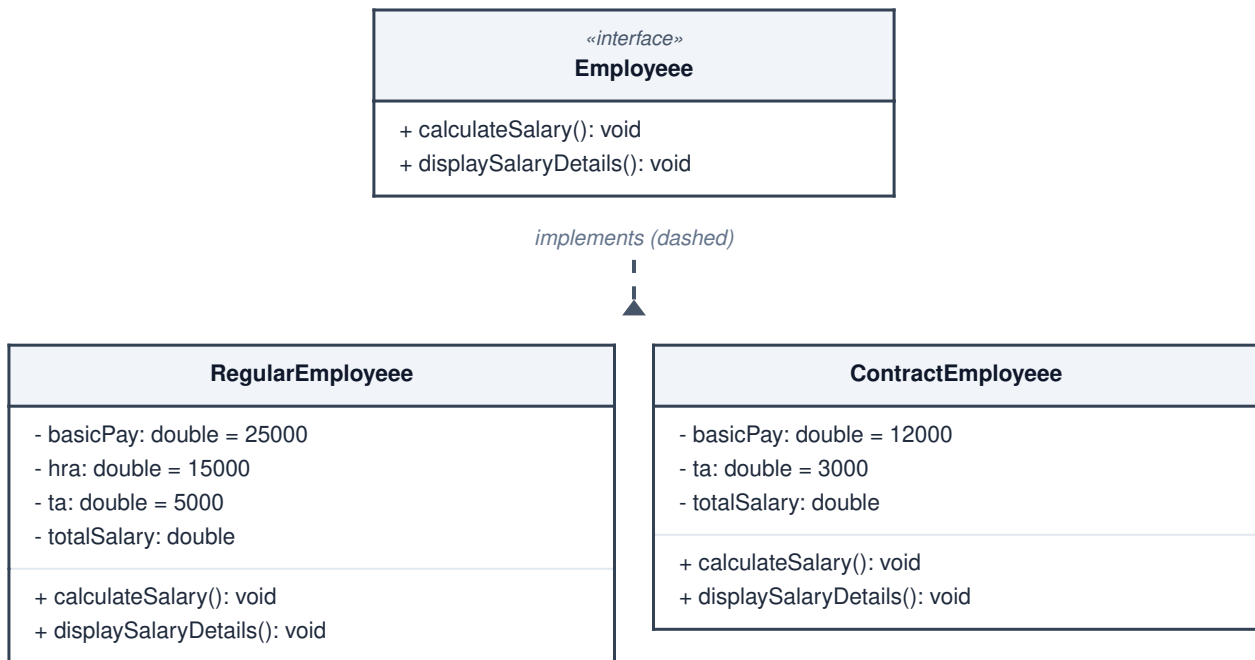
- **Regular Employees:** Basic Pay = ₹25,000 | HRA = ₹15,000 | TA = ₹5,000
- **Contract Employees:** Basic Pay = ₹12,000 | TA = ₹3,000 | HRA = ₹0 (No HRA)

Write a program to calculate and display the monthly salary details based on Employee ID input prefix (R for Regular, C for Contract).

## 2. STEP-BY-STEP IMPLEMENTATION PROCESS

- 1 **Define Interface (Employeee):** Declare abstract methods `calculateSalary()` and `displaySalaryDetails()` to create a common contract for all employee types.
- 2 **Implement Classes:** Create `RegularEmployeee` and `ContractEmployeee` implementing the `Employeee` interface, providing specific basic pay, allowances, and total salary computation logic.
- 3 **Access via Interface Reference:** In the main class `PayrollSystem`, declare an interface reference variable (`Employeee emp`) and dynamically bind it to concrete objects based on user input ID.
- 4 **Execute Runtime Polymorphism:** Call `emp.displaySalaryDetails()` via the interface reference variable to output employee salary details dynamically.

### 3. UML DIAGRAM



### 4. JAVA CODE IMPLEMENTATION

```
package record;

import java.util.Scanner;

// 1. Defining the Interface
interface Employee {
    void calculateSalary();
    void displaySalaryDetails();
}

// 2. Regular Employee class implementing Employee interface
class RegularEmployee implements Employee {
    private double basicPay = 25000;
    private double hra = 15000;
    private double ta = 5000;
    private double totalSalary;

    @Override
    public void calculateSalary() {
        totalSalary = basicPay + hra + ta;
    }

    @Override
    public void displaySalaryDetails() {
        calculateSalary();
        System.out.println("Salary Details:");
        System.out.printf("Basic Pay: %.0f HRA: %.0f T.A: %.0f Total Amount: %.0f\n",
            basicPay, hra, ta, totalSalary);
    }
}

// 3. Contract Employee class implementing Employee interface
```

```

class ContractEmployee implements Employee {
    private double basicPay = 12000;
    private double ta = 3000;
    private double totalSalary;

    @Override
    public void calculateSalary() {
        totalSalary = basicPay + ta;
    }

    @Override
    public void displaySalaryDetails() {
        calculateSalary();
        System.out.println("Salary Details:");
        System.out.printf("Basic Pay: %.0f T.A: %.0f Total Amount: %.0f\n",
            basicPay, ta, totalSalary);
    }
}

// 4. Main class accessing implementations through interface references
public class PayrollSystem {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter Employee Id: ");
        String empId = scanner.nextLine().trim();

        // Interface reference variable
        Employee emp = null;

        // Determine employee type based on ID prefix
        if (empId.toUpperCase().startsWith("R")) {
            emp = new RegularEmployee(); // Reference pointing to Regular object
        } else if (empId.toUpperCase().startsWith("C")) {
            emp = new ContractEmployee(); // Reference pointing to Contract object
        } else {
            System.out.println("Invalid Employee ID Type.");
            scanner.close();
            return;
        }

        // Polymorphic call using interface reference
        emp.displaySalaryDetails();

        scanner.close();
    }
}

```

## 5. SAMPLE EXECUTION & OUTPUT

### TestCase 1: Regular Employee Input

#### CONSOLE OUTPUT

```

Enter Employee Id: R101
Salary Details:
Basic Pay: 25000 HRA: 15000 T.A: 5000 Total Amount: 45000

```

## TestCase 2: Contract Employee Input

### CONSOLE OUTPUT

Enter Employee Id: C202

Salary Details:

Basic Pay: 12000 T.A: 3000 Total Amount: 15000